

Neonatal Brain MRI Segmentation

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Abstract. The study focuses on semantic segmentation of neonatal brain images obtained by magnetic resonance imaging. Deep neural networks for semantic segmentation were selected, implemented, or existing implementations were utilized. The networks were used on available datasets from the iSeg 2019 and BONBID-HIE 2023 challenges and also our internal dataset TAČR 6-HIE to perform 4 segmentation tasks in total. The networks were evaluated using selected metrics and are presented in this study.

Keywords: Neonatal MRI · Semantic segmentation · Computer Vision · Medical image analysis · Hypoxic Ischemic Encephalopathy · HIE.

1 Introduction

Magnetic resonance imaging is a noninvasive technique for examining tissues in (living) organisms. The examination is recommended for newborns with neurological disorders, suspected brain damage, injuries, or malformations (e.g., hypoxic-ischemic encephalopathy), seizures, or head trauma [2, 6, 3]. Images can be used to detect and examine extent and location of brain abnormalities and accurate segmentation might be helpful. The imaging and segmentation is more challenging than in adults due to lower resolution, low contrast, and unclear tissue contrast distribution. Around 6 months of age, MRI voxel intensities of white and gray matter overlap significantly, making the segmentation even harder. As a result, segmentation tools developed for adults often perform poorly on infant brain images [11]. As a result specialized tools have been (or are actively) developed and deep neural networks for semantic segmentations can be utilized.

2 Datasets

Datasets from mentioned challenges were utilized and also internal dataset TAČR 6 - HIE which is being gathered in collaboration with Faculty hospital of Ostrava. Usage of the anonymized data in this study is compliant with the ethics.

iSeg 2019 contains processed images of infants aged around 6 months and the segmentation is classifying voxels as white matter, dark matter and cerebrospinal fluid. We use subjects for which two images and ground truth mask is available. More details for the dataset can be found out in article [11].

BONBID-HIE 2023 dataset [3] contains images of infants diagnosed with hypoxic-ischemic encephalopathy (HIE), and the task is to segment whether the voxel contains tissue damaged by HIE or not. The images have different sizes and voxel sizes and for processing the images are resampled to unified size of $256 \times 256 \times 64$ with $1\text{mm} \times 1\text{mm} \times 1\text{mm}$ voxel size. 2 images and one ground truth mask per subject is available. The amount of the brain tissue affected is different across the images.

The TAČR 6 - HIE dataset contains images of 1 to 2 day old newborns diagnosed with HIE. The images were captured by Siemens Prisma 3 tesla magnetic resonance using 32 channel head coil. The images were preprocessed using N4 bias correction, skull stripping and anonymized. After that, ground truth anatomical segmentation maps created using FreeSurfer infant pipeline [12] were created. Images for 20 subjects were provided for writing of this study.

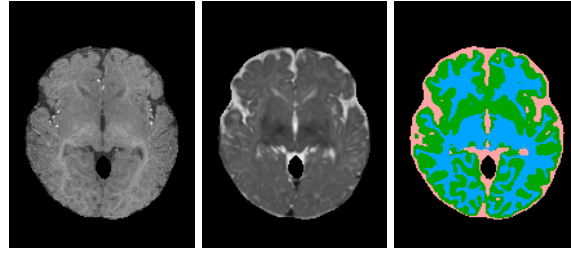
Table 1: Overview of datasets

Parameter	iSeg 2019	BONBID-HIE 2023	TAČR 6 - HIE
Number of subjects	10	81	20
Images per subject	2	2	1
Image size	$256 \times 160 \times 144$	various	$256 \times 256 \times 256$
Number of classes	4	2	31, 98

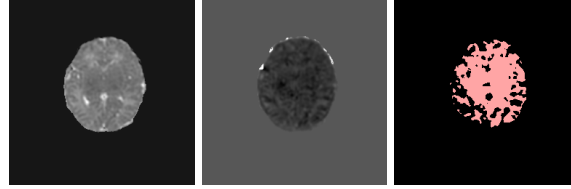
All the datasets have imbalanced classes, background is most prevalent class in all of them, but also the number of voxels representing brain tissue classes is imbalanced. Class imbalance is typical for medical image analysis and should be taken into account [9]. Example slices of images for datasets are in figure 1.

3 Methodology

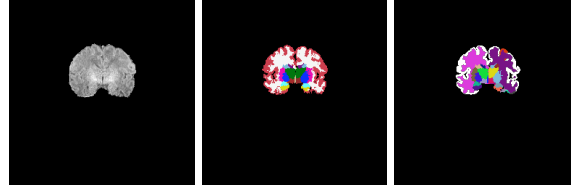
Same training, validation and testing images were used for training and evaluating every model. The datasets were split for training, testing and validation parts in ratio 80:10:10. Models were evaluated using metrics. We opted for metrics for binary segmentation, namely DICE coefficient, Hausdorff distance, Mutual information, Mahalanobis distance and Volumetric similarity for evaluating models. DICE coefficient was chosen as it is one of the most used metric [1] for evaluating semantic segmentation. Hausdorff distance and Mahalanobis distances were chosen from distance based metrics. Volumetric similarity was chosen because it measures the similarity from the point of volumes of segmented segments. Mutual information was chosen because it quantifies the amount of shared information between the predicted segmentation and the ground truth, making it effective for evaluating alignment and overall structural similarity. Definitions of metrics and utilized tool were also described in mentioned article [1]. For multiclass segmentations such as in case of iSeg 2019 and TAČR 6 - HIE segmentation tasks, weighted average of individual classes metrics were calculated



(a) iSeg 2019 subject (T1, T2 and segmentation map)



(b) BONBID-HIE 2023 subject (ADC, Z_ADC and segmentation map)



(c) TAČR 6 - HIE subject (T1, aseg and aseg+aparc segmentation maps)

Fig. 1: Examples of datasets images.

to get aggregated results for whole images. Relative volumes of classes in ground truth segmentation masks were used as corresponding weights.

4 Selected Models

We implemented convolutional neural networks based on architectures such as U-net [10] and DeepLab version 3 [4]. From existing implementations, we opted for nnUNet version 2, which is an automated tool developed for the semantic segmentation of medical images using the U-net architecture [8]. We compared both 2D and 3D full resolution configurations for all datasets. The second tool used was U-mamba, which is based on nnUnet which utilizes sequence state space based architecture implementation called Mamba [5]. Similarly 2D and 3D full resolution configurations, which use Mamba block only within the network's bottleneck were used. Source codes are available in repository [7].

5 Results

We trained, tested and evaluated models mentioned in section 4 using methodology specified in section 3. Table 2 contains medians and IQRs of metrics calculated for testing subjects of dataset BONBID-HIE 2023, table 3 for iSeg 2019, tables 4 and 5 for TAČR 6 - HIE datasets. Images 2, 3, 4 and 4 show visual comparison of one slice of resulting segmentations for testing subjects of respective datasets and ground truth. Incorrectly segmented voxels have got dark red colour. Next list briefly describes the results for each segmentation task.

- According to the results in table 2, the nnUNet (3D) model achieved the highest median DICE similarity in the BONBID-HIE 2023 test dataset. The 2D configuration reached a higher MI value with a smaller interquartile range. The custom implementation of the DeepLab v3 model achieved a lower median DICE similarity with a larger interquartile range. The smallest Hausdorff distance was obtained by the U-Mamba model with the 3D bottleneck configuration, while DeepLab v3 showed a higher median with smaller IQR. The highest volume similarity and Mahalanobis distance were reached by the DeepLab v3 model.
- According to the table 3, the nnUNet (3D) model achieved the highest DICE similarity and MI on the iSeg 2019 test dataset. The 2D configuration of nnUNet achieved the smallest Mahalanobis distance. The highest volume similarity was reached by the U-Mamba model 2D bottleneck configuration. The smallest Hausdorff distance was achieved by the 3D bottleneck configuration.
- According to the results in table 4, the custom implementation of U-Net (3D) achieved the highest median DICE similarity, the lowest median Hausdorff distance, the highest MI value, and the lowest median MHD on the TAČR 6-HIE (aseg) test dataset. Only the U-Mamba 3D bottleneck configuration reached a higher volume similarity.
- According to the results in table 5, the custom implementation of U-Net (3D) also achieved the highest median DICE similarity, the highest MI value, and the lowest median MHD in the TAČR 6-HIE (aseg+aparc) test dataset. The nnUNet in the 3D configuration achieved a lower Hausdorff distance but with a larger interquartile range. Only the U-Mamba model in the 3D bottleneck configuration achieved a higher volume similarity.

6 Conclusion

We implemented and utilized existing neural networks for four segmentation tasks. As the results show our implementations and configurations have not achieved satisfactory results for the BONBID-HIE 2023 dataset, where on selected data split we achieved maximum median DICE of only around 0.4.

Our implementation of U-net with 3D convolutions achieved worst score of approximately 0.83 in iSeg 2019 segmentation task. On the other hand, similar

Table 2: Medians and IQR of achieved metrics in BONBID-HIE 2023 segmentation. Maximal medians of similarities and mutual information and minimal medians of distances are highlighted.

Model	DICE	HD	VS	MI	MHD
U-net (3D)	0,000 (0,015)	162,123 (43,861)	0,017 (0,325)	0,000 (0,000)	5,858 (66,481)
nnUnet (2D)	0,087 (0,363)	49,123 (36,847)	0,213 (0,504)	0,002 (0,010)	0,953 (0,617)
nnUnet (3D)	0,408 (0,190)	32,997 (16,728)	0,671 (0,418)	0,002 (0,011)	0,631 (0,841)
DeepLab (v3)	0,407 (0,299)	31,443 (14,896)	0,740 (0,125)	0,001 (0,008)	0,481 (0,444)
U-Mamba (2D bot)	0,064 (0,407)	51,862 (41,881)	0,190 (0,780)	0,002 (0,010)	0,915 (0,665)
U-Mamba (3D bot)	0,168 (0,343)	30,564 (22,293)	0,586 (0,254)	0,001 (0,011)	0,941 (0,831)

Table 3: Medians and IQR of achieved metrics in iSeg 2019 segmentation. Maximal medians of similarities and mutual information and minimal medians of distances are highlighted.

Model	DICE	HD	VS	MI	MHD
U-net (3D)	0,825	70,494	0,973	0,165	0,038
nnUnet (2D)	0,901	7,730	0,973	0,200	0,012
nnUnet (3D)	0,921	7,296	0,965	0,210	0,015
DeepLab (v3)	0,834	7,575	0,932	0,170	0,024
U-Mamba (2D bot)	0,894	6,618	0,975	0,196	0,015
U-Mamba (3D bot)	0,914	5,838	0,960	0,207	0,020

Table 4: Medians and IQR of achieved metrics in TAČR 6 - HIE aseg segmentation. Maximal medians of similarities and mutual information and minimal medians of distances are highlighted.

Model	DICE	HD	VS	MI	MHD
U-net (3D)	0,751 (0,012)	20,546 (2,143)	0,929 (0,013)	0,021 (0,001)	0,200 (0,054)
nnUnet (2D)	0,635 (0,083)	40,181 (0,383)	0,980 (0,003)	0,015 (0,003)	0,733 (0,230)
nnUnet (3D)	0,463 (0,438)	23,678 (18,092)	0,983 (0,004)	0,013 (0,013)	1,999 (1,953)
DeepLab (v3)	0,066 (0,022)	63,642 (1,331)	0,766 (0,169)	0,001 (0,000)	4,266 (0,285)
U-Mamba (2D bot)	0,673 (0,086)	40,393 (0,570)	0,981 (0,005)	0,017 (0,003)	0,616 (0,258)
U-Mamba (3D bot)	0,459 (0,434)	23,832 (17,829)	0,985 (0,005)	0,013 (0,013)	1,998 (1,947)

Table 5: Medians and IQR of achieved metrics in TAČR 6 - HIE aseg+aparc segmentation. Maximal medians of similarities and mutual information and minimal medians of distances are highlighted.

Model	DICE	HD	VS	MI	MHD
U-net (3D)	0,645 (0,010)	24,705 (2,323)	0,872 (0,006)	0,014 (0,000)	0,448 (0,066)
nnUnet (2D)	0,577 (0,116)	34,658 (2,075)	0,948 (0,012)	0,009 (0,003)	0,885 (0,418)
nnUnet (3D)	0,423 (0,399)	21,813 (16,192)	0,957 (0,010)	0,007 (0,007)	3,060 (2,896)
DeepLab (v3)	0,045 (0,007)	66,577 (3,306)	0,646 (0,186)	0,001 (0,000)	6,997 (0,785)
U-Mamba (2D bot)	0,150 (0,011)	29,035 (0,894)	0,316 (0,012)	0,004 (0,001)	1,718 (0,151)
U-Mamba (3D bot)	0,416 (0,392)	21,858 (16,168)	0,957 (0,009)	0,007 (0,007)	3,053 (2,880)

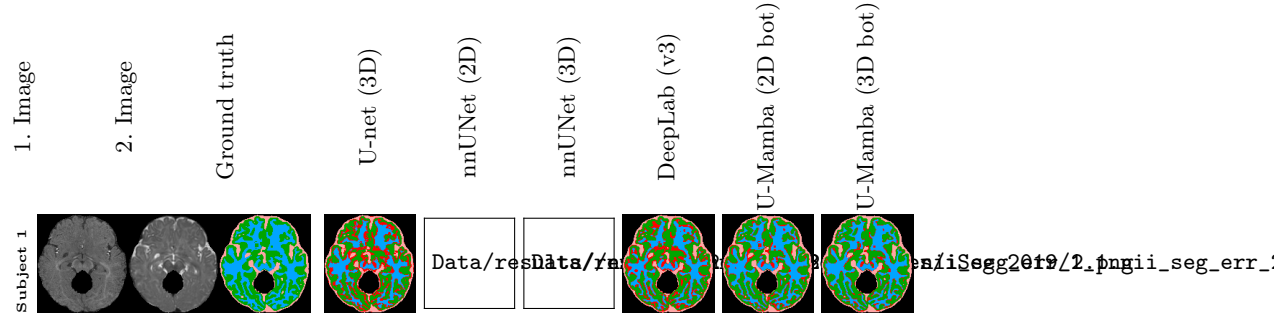


Fig. 2: Visual comparison of slices of images, ground truth and segmentations created by models for iSeg2019 segmentations. Incorrectly classified pixels are dark red.

Subject	Number of regions	Brain volume percentage
1	0	0 %
2	14	0,98 %
3	27	8,32 %
4	2	0,06 %
5	2	0,035 %
6	20	0,44 %
7	1	0,037 %
8	3	59,31 %
9	117	1,44 %

Table 6: Overview of number of regions and brain volume percentage for test subjects of BONBID-HIE 2023 dataset.

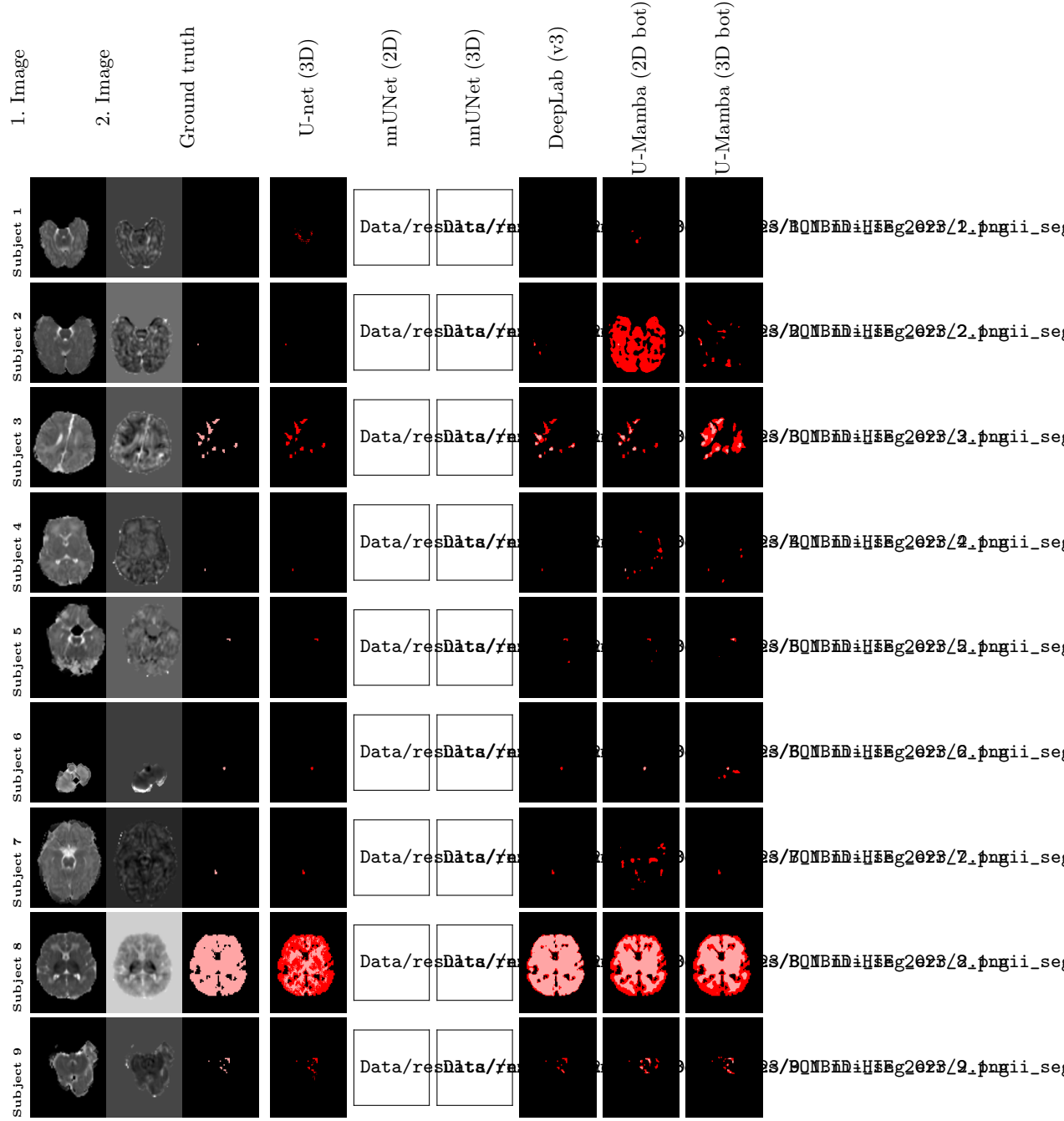


Fig. 3: Visual comparison of slices of images, ground truth and segmentations created by models for BONBID-HIE 2023 segmentations. Incorrectly classified pixels are dark red. Legend for subjects is in table 6.

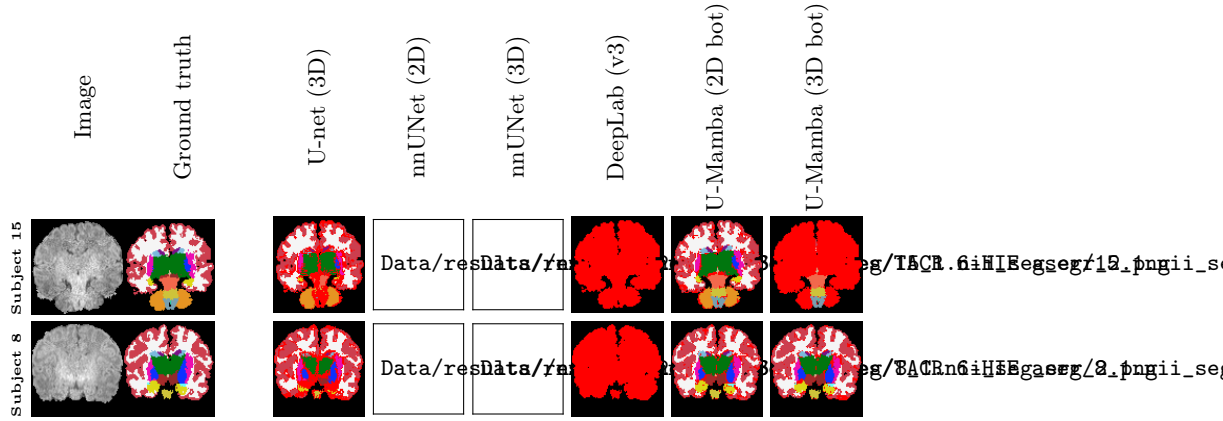
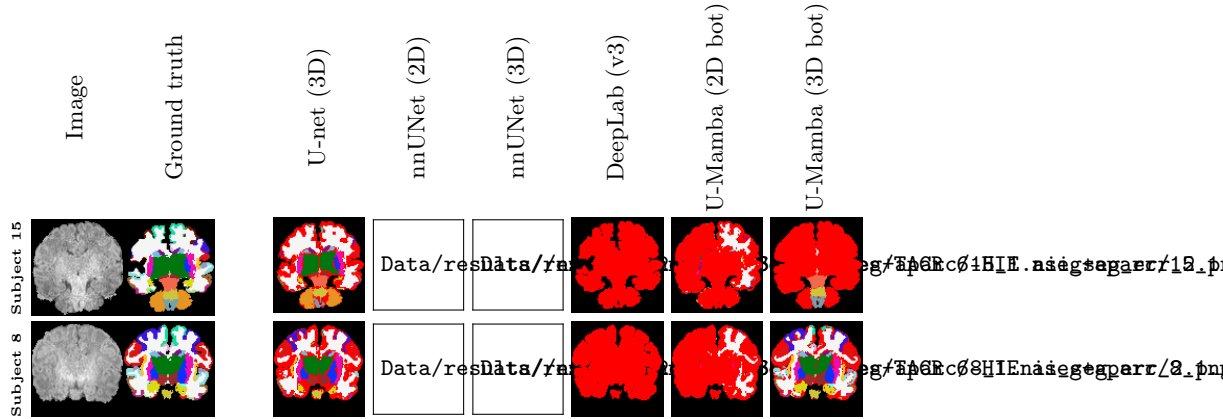


Fig. 4: Visual comparison of slices of images, ground truth and segmentations created by models for TAČR 6-HIE (aseg) segmentations. Incorrectly classified pixels are dark red.

Fig. 5: Visual comparison of slices of images, ground truth and segmentations created by models for TAČR 6-HIE (aseg+aparc) segmentations. Incorrectly classified pixels are dark red



network architecture configured by nnUnet achieved best DICE score (around 0.92). In both segmentation tasks of TAČR 6 - HIE dataset our implementation achieved the best scores (around 10% better than nnUnet). This result is quite surprising, as we were not expecting to surpass nnUnet with such a simple architecture without extended fine tuning.

Overall the results are not satisfactory for clinical application and further research is needed. One limitation in our methodology which should be addressed is the lack of cross validation, which would provide more robust evaluation of models in the case of such small datasets, which were examined.

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